Note from Gunnar (1 May 2026): RG of DP — exponents without explicit loops
transcribed; reply lives in `reply.tex`

Preamble (Gunnar, in his own words)

I want to understand the claim of "exponents without loop integrals" a bit better.

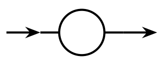
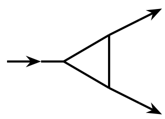
In the case of Doi–Peliti I can see how this may work. But Claude seemed to suggest that in the Gribov process (which is directed percolation) exponents are a matter of pure combinatorics, which I don't see.

So now I will derive the exponents of directed percolation and it will become clear that you get them without calculating integrals explicitly, although you need to know some of their behaviour. So yes: ***exponents without explicit loops, but no, nothing to do with combinatorics****.*

As an aside, I wonder whether analytic combinatorics can help characterising the wild forest of diagrams you get when you attempt to write down φ^4-theory in Doi–Peliti using a Hermite basis.

DP — to one loop

To one loop there are two diagrams to be considered:


and


Bubble (self-energy). With symmetry factor $\frac{1}{2}\cdot 2 = 1$,

$$
\begin{aligned}
\text{(bubble)} \;&=\; 2g^2 \int \frac{d\omega'}{2\pi}\frac{d^dk'}{(2\pi)^d}\; \frac{1}{-\mathrm{i}(\omega+\omega') + D(k+k')^2 + r} \cdot \frac{1}{\mathrm{i}\omega' + Dk'^2 + r} \\
&=\; 2g^2 \int d^dh' \; \frac{1}{-\mathrm{i}\omega + D(k+h')^2 + D(k-h')^2 + 2r}\bigg|_{h'=k'+k/2} \\
&\qquad \text{using } D(k{+}k'/2)^2 + D(k{-}k'/2)^2 \;=\; 2Dk'^2 + Dk^2/2.
\end{aligned}
$$

After the standard Schwinger / Feynman parameter and a d-dimensional angular integration,

$$
\begin{aligned}
\text{(bubble)} \;&=\; g^2\,K_d \int_0^\infty ds\; \frac{s^{\,d/2-1}}{-\mathrm{i}\omega + 2Ds + Dk^2/2 + 2r} \\
&=\; \frac{g^2}{2D}\,K_d \left(\tfrac{-\mathrm{i}\omega + Dk^2/2 + 2r}{2D}\right)^{d/2-1} \frac{\Gamma(1-d/2)\,\Gamma(d/2)}{\Gamma(1)} \\
&=\; \frac{g^2}{(2D)^{d/2}}\,K_d\,\Gamma(d/2)\left(-\mathrm{i}\omega + Dk^2/2 + 2r\right)^{d/2-1} \frac{\Gamma(\varepsilon/2)}{1-d/2} \\
&\equiv\; 2g^2\,\mathcal{I}_{-O-}(k,\omega,r) \;=\; 2g^2\,A\,\Gamma(\varepsilon/2)\,\frac{\left(-\mathrm{i}\omega + Dk^2/2 + 2r\right)^{d/2-1}}{1-d/2}, \quad A \;\equiv\; \frac{K_d}{D^{d/2}}.
\end{aligned}
$$

Triangle (vertex correction). Symmetry factor $\frac{1}{2} \cdot 16 = 16$, and the key trick: $-\partial_r$ acting on a propagator doubles it, $\partial_r\big[(\cdot + r)^{-1}\big] = -(\cdot + r)^{-2}$, so

$$(\text{triangle}) \;=\; -16\, g^3 \left(-\tfrac{1}{2}\partial_r\right) \mathcal{I}_{-O-}(0,0,r) \quad [\text{idea: } -\partial_r \text{ "doubles up" a propagator}].$$

Tree level multiplicity is 2, giving

$$(\text{triangle}) \;=\; +16\, g^3\, A\, \Gamma(\varepsilon/2)\, (2r)^{d/2-2}.$$

Effective coupling. Define

$$\bar g \;=\; \frac{2g^2 A}{D^{d/2}} (2r)^{-\varepsilon/2} \;=\; 2\,\frac{g^2}{D^2}\, A \left(\tfrac{2r}{D}\right)^{-\varepsilon/2}.$$

Z-factor relations

Reading off the four divergences (the colour highlights match Gunnar's pen):

$$\begin{aligned} Z_r\, Z_\phi\, Z_{\tilde\phi} &= 1 + 2\,\bar g, \\ Z_\phi\, Z_{\tilde\phi} &= 1 + \bar g, \\ Z_D\, Z_\phi\, Z_{\tilde\phi} &= 1 + \tfrac{1}{2}\bar g, \\ Z_g\, Z_\phi\, Z_{\tilde\phi}^2 &= 1 + 4\,\bar g \quad \big[\text{from } \tfrac{16}{2\cdot 2}\big]. \end{aligned}$$

Solving with the Reggeon Ward identity $Z_\phi = Z_{\tilde\phi}$:

$$Z_r \;=\; 1 + \bar g, \qquad Z_D \;=\; 1 - \tfrac{1}{2}\bar g, \qquad Z_\phi \;=\; Z_{\tilde\phi} \;=\; 1 + \tfrac{1}{2}\bar g, \qquad Z_g \;=\; 1 + \tfrac{5}{2}\bar g.$$

(Equivalently $Z_{\bar g} = 1 + \big(2 \cdot \frac{5}{2} + 2 \cdot \frac{1}{2}\big)\bar g = 1 + 6\bar g$.)

β-function and exponents

From $Z_{\bar g} = 1 + 6\bar g$:

$$\beta_{\bar g} \;=\; -\varepsilon\,\bar g \;+\; 6\,\bar g^2 \quad\Longrightarrow\quad \bar g^* \;=\; \tfrac{\varepsilon}{6}.$$

The anomalous dimensions evaluate to

$$\gamma_r \;=\; \tfrac{\varepsilon}{6}, \qquad \gamma_D \;=\; -\tfrac{\varepsilon}{12}, \qquad \gamma_\phi \;=\; \tfrac{\varepsilon}{12},$$

and the standard Tauber relations give

$$\boxed{\frac{1}{\nu} \;=\; 2 - (\gamma_r - \gamma_D) \;=\; 2 - \tfrac{\varepsilon}{4}, \qquad z \;=\; 2 + \gamma_D \;=\; 2 - \tfrac{\varepsilon}{12}, \qquad \eta \;=\; -2\gamma_\phi - \gamma_D \;=\; -\tfrac{\varepsilon}{12}.}$$

These are the exponents to one loop.

Sash's commentary → `reply.tex` in this folder.